Jonathan Bong Lin Choon

Singapore | jonbong@gmail.com
://linkedin.com/in/jonathan-bong-5a229840/ | ://github.com/jon-bong

Professional Summary

Over 26 years as an I.T professional in the solutions provider, telecommunication, system integration, data-collection, lottery, digital advertising and energy industries; specialising in the areas of software development, testing and documentation, operations and end-user support. Developed new software systems, enhanced existing systems for the education, public sector, airline, financial, lottery (gaming), digital advertising and energy industries. Bachelor of Computing with a double major in systems development and business systems.

Skills

Core Ecosystem: Microsoft .NET | C#

Web Frameworks: ASP.NET | ASP.NET MVC

API & Integration: ASP.NET Web API | RESTful APIs | JSON | XML | XSD

Databases: Microsoft SQL Server | MySQL | SQLite

Tools & Cloud: Html Agility Pack | Selenium Web Driver | Git/GitHub | Sandcastle |
Microsoft Dynamics 365 Finance and Operations |
Microsoft Azure (Vision in Foundry Tools, Data Explorer, SQL, SWA, Storage) |
Amazon Web Services (EC2, S3, RDS, DMS)

Professional Experience

Sembcorp Utilities Services Pte Ltd (Sembcorp Industries Ltd) **Sep 2018 – Apr 2026**
Application Operations Specialist

Contributed to the organisation’s digital transformation initiatives in addition to enhancement, maintenance, support and operations of the energy billing and renewables systems.

- New Billing System
  - Collaborated with the project team to execute data migration of Sembcorp Power's residential customers from the outgoing vendor system to the new platform.
  - Orchestrated data configurations and migration of Sembcorp's integrated utilities customer and pricing data, establishing pricing formulae in the new system as part of the digitisation initiatives from current Excel-dependent processes.
  - Developed business models of the current Excel-dependent Solar pre-billing process, preparing for data modelling, subsequent migration, and setup of Solar Billing related data within the new system.

- Legacy Billing System
  - Managed support and operations for the legacy billing system, ensuring system uptime, providing software support, and executing ad hoc billing tasks to maintain optimal system performance.
  - Addressed maintenance and corrective work, resolving issues reported by Billing and Finance stakeholders.
  - Refactored and rewrote in-house programs to alleviate application inefficiencies and developed new programs to meet the Billing team's evolving requirements, including integrations with external payment vendor systems like AXS and MasterCard Payment Gateway System (MPGS).
  - Designed solutions to allow the system to interact with modern systems to allow the exchange and processing of billing-related data with the company's enterprise and cloud systems.
- Solar (Renewables) Systems
  - Worked closely with the Energy Management Systems Division of Singapore's Energy Market Authority (EMA) for the reporting of photovoltaic energy received from commercial solar sites, and also performing transmission tests of photovoltaic data files prior to new site commissions.
  - Supported “Business as Usual” (BAU) operations for Sembcorp Solar Singapore systems, ensured up-to-date and timely receipt of daily Solar measurement data files from commercial sites for monthly downstream billing.
  - Worked closely with the internal customer management team to ensure accurate Solar export data from the company’s “SolarNova” projects with HDB was downloaded from Singapore Power's File Exchange system to facilitate downstream billing.
  - Revamped the extraction mechanism for Solar export data and charges from Town Councils' digital bills by leveraging Microsoft Azure (AI) Vision in Foundry Tools, ensuring successful capture of all billing charges and usage data for downstream billing.

Smartidea Pte Ltd **Sep 2014 – Jun 2018**
Software Developer

Involved in R & D (including proof-of-concept), development, operations, support and maintenance of new/existing systems related to digital advertising.

- Strategic Development Projects
  - Worked closely with the technical and strategic-planning teams to create a software platform for the strategic-development team to keep better track of the company’s new game initiatives, downloads and operational statistics which in turn enable the sales team to come out with effective strategies and effective business proposals to the company’s clients.
- Web Payment Services
  - Successfully designed and implemented a company-wide payment system that interacted with new and existing payment gateways offered by the company's merchant banks, thereby allowing various in-house systems of varying .NET programming frameworks to make successful payment-related operations.

- Proof-of-Concept Research Modules
  - Participated in the company's research modules, often consisting of proof-of-concept or investigative work to allow management and stakeholders to conduct and complete feasibility study before deciding to embark on keystone projects.
- KeywordSpy (SEO Commercial Product)
  - In charge of back-end operations (periodic batch-processing, monitoring and routine database-related activities) of the company's flagship product by being on operational standby during and after official working hours, usually reacting to notification alarms triggered by third-party monitoring systems.
  - Involved in enhancement and maintenance of back-end and sub-systems as well as front-end web applications (main product and billing modules in ASP.NET), often leveraging on Amazon Web Services (AWS) technologies.
  - Provided end-user support for internal Finance users and business customers.
  - Administered the various staging and live MySQL databases.
  - Developed utility programs to aid in the daily operations, leveraging on the Amazon Web Services (AWS) Software Development Kit (SDK) for Microsoft .NET projects.
  - Successfully migrated the system's back-end 5 application/database servers from a traditional server-hosting environment to Amazon Web Services.

Singapore Pools (Private) Ltd **Jan 2012 – Sep 2014**
Applications Analyst

Involved in project management, development maintenance and support of new and existing application systems related to the company's lottery operations.

- Draw Operations Management Systems
  - Took over historical backlog of 10 enhancement requests and completed the batch within 2 months while facing a steep learning curve of the unfamiliar software environment.
  - Provided end-user support during the company's live lottery draws.
  - Worked closely with the system owner during the relocation of company premises and company-wide disaster recovery exercise (planning, preparation, configuration etc) to ensure all system-related equipment, hardware & software operations were in working condition throughout the entire relocation process.
- Video Streaming Server System
  - Ensured all preceding versions of source code & binaries were appropriately tagged, updated and archived into the department's repository; updated related documentation to reflect most-recent changes to the system.
  - Undertook all enhancement works related to the changes in lottery prize structures of the company's lottery products.
  - Provided end-user support during the system's operational hours.

- Sweep Validation System
  - Collaborated with the company's IT vendor (Accenture Singapore) in the SIT and UAT phases during the recent system revamp, providing the vendor with crucial information and managing the expectations of the business users in the testing phases.
  - Took over ownership of the system after "go-live"; currently undertaking all documentation, system enhancements, system build and support work.
  - Worked closely with the department's SAP team to provide useful interfaces required by the team for their financial projects.
- Secured Document System
  - Worked together with the company's Operations Support department to re-establish their existing secured document repository from ground up; converted and enhanced all related programs previously implemented using different frameworks (PHP, JAVA) to the Microsoft .NET framework to keep in line with the company's IT direction.
- Pre-Paid Account Branch Application
  - Involved in the enhancement of the related to the changes in lottery prize structures of the company's lottery products.
  - Provided end-user support during the sales operational hours of the company's branches throughout the island.

NCS Group **Jan 2009 – Mar 2011**
Consultant, Application

Involved in development and maintenance projects in the government and education sectors.

- Temasek Polytechnic Student and Management System
  - Performed regression and white-box testing for enhancement and corrective works before deployment to the UAT/SIT environments.
  - Provided end-user support for the system.
  - Worked closely with the Student Information Systems department to obtain user requirements for desired enhancements to be made on the system; subsequently converted requirements to design specifications for subsequent development work.
- Human Resource Management System (Ministry of Foreign Affairs)
  - Member of the pioneer development team; performed a thorough study of the existing legacy system with regards to data structure (existing database tables, un-normalised data).
  - Developed a data migration program to validate and process 200 data files of over 70 different formats bearing sensitive information into the system's database to be utilised by other sub-modules of the system.
  - Performed duplication and data redundancy checks prior to data migration in the legacy system; successfully accounted for all data records and retained only unique records during the data migration phase from the legacy database to the new system database.

- Claims and Financial Budget Forecast System (Ministry of Foreign Affairs)
  - Played an end-user support role for financial systems used by ministry officers by attending to them face-to-face to rectify issues encountered
  - Involved in corrective/enhancement work (such as creating and fine-tuning drill-down SQL queries for claims and financial reports, front-end ASP.NET websites and back-end batch jobs) while completing them within targeted project deadlines and stringent SLA periods.
- Integrated Manpower Survey System (Ministry of Manpower)
  - Part of the Quality Assurance team that performed regression testing on the system.
  - Gathered and reviewed feedback and requirements from end-users; converted information to technical specifications for use by the team developers in their development work.

Singapore Computer Systems (SCS) Ltd (NCS Group) May 2008 – Dec 2008
Application Specialist

Involved in software testing, documentation, deployment activities, database administration and server maintenance work for financial and transport-related projects.

- Marketing and Relationship Management System (Singapore Airlines Ltd)
  - Developed test cases and performed regression testing on enhancement/maintenance work performed by the team developers.
  - Worked closely with Singapore Airlines' IT division and other international airline carriers during the UAT of the “Frequent Flyer” module to ensure that input and output files were correctly generated in accordance to the “Partnership Accrual Interface” specification.
  - Modified and updated system documentation (functional specifications, technical design specifications and test plans) after every work sign-off.
  - Performed server-related duties such as (Oracle 9i) database administration (backup, script executions, database objects access rights etc), middle-tier (BEA Weblogic) server maintenance to support fellow developers in the project team.
  - Undertook software build & deployment activities to the UAT & live environments to meet both scheduled & ad-hoc requests made by the end user.
- DealMaker New Financial Model (VISA Asia Pacific)
  - Worked together with the team lead on the system documentation (functional and technical specifications).
  - Performed regression testing on the developed modules based on the test cases derived from the written specifications.
  - Worked closely with Visa's business analysts during the UAT, SIT & Staging periods on user feedback and change requests until the application was deployed and signed off.

Alcatel-Lucent Enterprise - Singapore ALE Pte. Ltd. **Mar 2007 – Apr 2008**
Software Developer

Involved in software loading/verification of the company's optical network equipment and peripherals, software development and computerization of a company department's in-house inventory system.

- Installed, configured and tested software in over 1000 sets of the company's optical network equipment bound for delivery to customers; successfully tested and delivered equipment to customers within the stipulated delivery schedules.
- Worked together with the technical manager & engineers in setting up network equipment for on-site user trainings and in-house problem-simulation & troubleshooting.
- Conducted user requirements gathering, analysis and design which subsequently led to the development of an inventory management system for use by staff to keep check on the current stock levels of the company's network equipment & peripherals.

A-Control Pte Ltd **Mar 2005 – Nov 2005**
Software Engineer

Performed design, implementation, deployment, enhancement and maintenance works on desktop PC software, embedded firmware and SCADA systems.

- Ultra-process Water (ST Microelectronics Pte Ltd)
  - Worked together with fellow sub-contractor company in developing a SCADA application that oversees the regulation of water & temperature within a wafer fabrication plant.
- AMK6e Alarm Notification System (ST Microelectronics Pte Ltd)
  - Refactored and enhanced an existing alarm notification system to eradicate memory leak & software crash issues.
  - Successfully enhanced software to seamlessly work with existing SCADA systems with a significant decrease in user intervention (from a daily to a fortnightly human intervention).
- Gas Monitoring System (ST Microelectronics Pte Ltd)
  - Developed and troubleshoot embedded programs such as protocol drivers and data converters for loading into gateway devices and Programmable Logic Controllers (PLCs).
  - Deployed 500 such devices over the wafer fabrication plant to integrate with existing SCADA monitoring application.

SATO Asia Pacific Pte. Ltd. **Apr 2002 – Aug 2002**
Programmer

Involved in the development of customised application programs and barcode software for data collection purposes, and also customer support for the company's main line of business.

- Developed customised PC desktop software based on customer requirements for use in conjunction with the company's proprietary barcode scanners and printers.

- Wrote user manuals and documentation of software programs developed for the customers.
- Provided first-level support (troubleshooting/maintenance services) to customers; addressed almost 80% of issues assigned while the remainder was escalated to the senior engineers.
- Developed an in-house meeting room reservation system for use by internal staff to aid in the daily administrative activities within the company.

Beca **Jul 2000 – Nov 2001**
Programmer

Involved in the design and development of the company's ERP system.

- Involved in the design and implementation of the company's in-house ERP system, in the form of a multi-tier client/server office document management system using Microsoft COM/COM+ technologies.
- Participated in the (object-oriented) analysis and design phase of the company's enterprise-level document management system via UML use case analysis & sequence diagrams.
- Developed data-access layer components (COM DLLs) for use by middle-tier business logic components in the COM+ environment.
- Worked together with the company's Senior Software Engineer in successfully creating add-ins for Microsoft Word automation to allow end-users to conveniently fill up electronic forms (ISO forms & tender documents) for submission via the company's internal workflow process.

Open Source & Personal Projects

MSSL EBT System Data Message Model and Interface Definitions **2026**
://jon-bong.github.io/mssl-ebt-models/
C#, XSD, XML Serialisation, .NET Standard, .NET Core, NuGet, Sandcastle, GitHub

An open-source energy market developer class library distributed via NuGet to streamline B2B integrations with Singapore's Open Electricity Market (OEM) infrastructure.

- Programmed domain-specific data models conforming to the Market Support Services Licensee (MSSL) Electronic Business Transaction (EBT) standards.
- Engineered a multi-targeted architecture natively supporting a vast range of environments, including legacy .NET Framework (3.5 to 4.8.1), .NET Standard (2.0/2.1), and modern .NET platforms (.NET 5.0 through .NET 10.0).
- Developed strongly typed processing structures for high-volume XML messaging and compressed Base64/ZIP/CSV utility billing telemetry.
- Leveraged interfaces defined in the package to enable energy sector developers to build custom data streaming and parsing workflows.
- Maintained public developer documentation and API reference materials via GitHub Pages and SandCastle.

EMC NEMS Data Services Model Definitions **2026**
://jon-bong.github.io/emc-sewplus-nems-models/
C#, XSD, XML Serialisation, .NET Standard, .NET Core, NuGet, Sandcastle, GitHub

An open-source .NET class library that provides strongly typed model definitions for data services within the National Electricity Market of Singapore (NEMS) system managed by the Energy Market Company (EMC). The library acts as a robust data wrapper based on the official NEMS Data Services Specification.

- Strongly Typed Architecture: Designed strongly typed classes and enumerations to handle EMC Download Reports and Submission messages, ensuring type safety and reducing runtime errors.
- Flexible Serialisation: Implemented XML serialisation and deserialisation capabilities, allowing developers to process real-time market data (like Uniform Singapore Energy Price, demand, and reserve data) using individual types or broad report categories.
- Cross-Platform Compatibility: Engineered the package to support a massive ecosystem of target frameworks, including .NET Standard 2.0/2.1, .NET Core 3.0 up to .NET 10.0, and legacy .NET Framework 4.0+.
- Automated Documentation: Generated comprehensive API reference guides using Sandcastle to lower onboarding friction for external developers.

Professional Courses

AZ-900 Microsoft Azure Fundamentals **2025**
WSQ ICT-DIT-4020-1.1 Cloud Computing-4
Tertiary InfoTech Pte. Ltd.

AI-900 Azure AI Fundamentals **2025**
WSQ ICT-TEM-3034-1.1 Artificial Intelligence Application in Product Development-3
Tertiary InfoTech Pte. Ltd.

MECOMS 365 Functional Overview **2024**
Ferranti Pte Ltd

Low-Powered Wide-Area Networks **2019**
School of Engineering, Nanyang Polytechnic

System Operations on AWS **2015**
Amazon Web Services

Microsoft Official Course 20486B: Developing ASP.NET MVC 4 Web Applications **2015**
Microsoft Official Course 20480B: Programming in HTML5 with JavaScript and CSS3
Microsoft Certified Application Developer (MCAD) on Microsoft .NET 1.0 **2006**
NetAssist Services Pte Ltd

Borland Silk Test, Borland Silk Performer **2008**
Borland Software Corporation

WonderWare InTouch 9.0 Basic **2005**
Servo Dynamics Pte Ltd

CREST Programme: National Critical Enabling Skills Training Programme (Listening and Oral Communication Module) **2001**
META HR & Communication Pte Ltd

JAVA Programming **1999**
Singapore Polytechnic

Education

Monash University **Jan 2004**
Bachelor of Computing: System Development/Business Systems

Singapore Polytechnic **Jun 2000**
Diploma in Computer Information Systems

Anderson Junior College **Dec 1994**
General Certificate of Education Advanced (GCE A-Level)

Anderson Secondary School **Dec 1992**
General Certificate of Education Ordinary (GCE O-Level)